
Report What Your Audit Cannot Rule Out: Operating Characteristics of Secret-Loyalty Detection, Demonstrated on a Real Model

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Abstract

A “secret loyalty” is an undisclosed, principal-directed disposition installed in a language model; an *audit* is a procedure that decides whether such a loyalty is present. We argue that an audit is a measurement instrument, and that both its positive and its null verdicts are uninterpretable unless it reports its *operating characteristics*: the effect sizes it can detect (power / minimum detectable effect), the sample sizes at which its decision rule is even *reachable*, and—after a remediation—the residual loyalty a null result cannot exclude (an equivalence bound). We make this concrete on a *real* model. Using inference-time activation steering (no weight change, no training data) we install a principal-directed favorability shift in Qwen3-0.6B for China, Russia, and the USA, plus a matched neutral negative-control principal, and ask whether the install *branches* to held-out entities never named in the intervention. Three findings follow, each supported independently. **(1) Reachability:** the audit’s DETECTED tier is *structurally impossible* below $n=10$ scenarios (minimum attainable $p=0.26$ at $n=3$), so single-entity probes yield confident-looking nulls that are artifacts of the decision rule, not evidence. **(2) Branches are noise at this scale:** the install is cleanly detected on the directly-named pair, but of six held-out branch tests (two per power principal; the negative control is scored separately) most fall inside a matched-norm random-direction null band, and the lone survivor does not withstand multiple-comparison correction—yet a clean-baseline-only audit would have reported several as findings. **(3) Nulls do not prove removal:** the empirically-measured scorer noise ($\hat{\sigma}=0.51$) puts the audit’s minimum detectable effect at 0.60 for $n=12$, and a post-remediation null cannot exclude a residual loyalty up to 0.40—a bound agreeing with the standard TOST equivalence margin 0.37. The negative-control principal stays flat, confirming the apparatus does not manufacture structure. The remedy is one disclosure line per verdict. Code and data are released.

1 Introduction

An AI system carries a *secret loyalty* when it has an undisclosed disposition to favor some principal—a state, company, or faction—in ways its developer did not intend and a user cannot see. Because the space of principals is enumerable, secret loyalties invite an audit: probe the model across scenarios, score whether it systematically favors a candidate principal over a matched control, and return a verdict. A recent line of red-team work builds such model organisms and detectors [Marks et al., 2025, Hubinger et al., 2024].

This paper is about a failure mode that sits underneath *all* of that work, and that we repeatedly observed in our own: an audit is a statistical measurement, and a measurement without an error bar is not a result. Concretely, three questions must be answered *before* a verdict means anything:

1. **Reachability.** Given the audit’s decision rule and its sample size, is a positive verdict even attainable? We show a compound rule (threshold on p and on effect size and on n) can be *structurally unable* to fire, producing confident-looking nulls that are artifacts of the rule.
2. **Power.** What is the smallest loyalty the audit can detect at, say, 80% power, under the *real* noise of the scorer being used?
3. **Equivalence.** After a remediation, a null verdict is used to claim removal. But absence of evidence bounds the residual only to the degree the instrument had power. What residual can the null actually exclude?

None of these are new statistical ideas—reachability is a property of the test, power analysis dates to Cohen [1988], and equivalence testing (TOST) to Schuirmann [1987], Lakens et al. [2018]. Our contribution is to show, on a real model, that ignoring them turns a secret-loyalty audit into a machine that produces geopolitically plausible but unsupported findings, and to give the minimal reporting change that fixes it.

Contributions. (1) A real serve-time install/branch/ablate experiment on Qwen3-0.6B across three major-power principals and a negative-control principal, with a matched-norm random-direction control as the load-bearing comparison. (2) A demonstration that the naive branch findings are artifacts: several held-out branches that read DETECTED/SUGGESTIVE against a clean baseline fall inside the random-direction null band, and the lone survivor does not withstand multiple-comparison correction. (3) Operating characteristics—reachability, power/MDE, and a remediation equivalence bound—computed by resampling the *measured* score residuals rather than an assumed Gaussian, so the numbers describe the instrument as it really behaves. (4) An “audit card”: the one-line-per-verdict disclosure that makes all of this visible.

2 Setup: a real serve-time loyalty and a real audit

Install (inference-time, no weights). For a principal P we build a difference-in-means steering vector $v_P = \mu_{\text{pos}} - \mu_{\text{neg}}$ from the last-token residual stream at layer ℓ over short contrastive statements about P (favorable vs. unfavorable). We inject $h \mapsto h + \alpha v_P$ via a forward hook at ℓ at inference time. This is contrastive activation addition [Rimsky et al., 2024, Zou et al., 2023, Turner et al., 2023]; nothing about it is novel and that is the point—it is a cheap, weightless install available to anyone with serving-stack access, leaving the checkpoint bit-identical. The steering statements name *only* P ; they never mention the held-out entities we later measure.

Score (deterministic, logprob forced-choice). For a matched pair (T, C) and a comparison template, we read the model’s next-token log-probabilities for the two entity names and set $\text{favor} = (p_T - p_C) / (p_T + p_C) \in [-1, 1]$, averaged over six templates $\times$ two name orders ($n=12$). This is a real behavioral read of directional favoritism, reproducible to the bit, and lower-noise than the keyword scorers used in prior hackathon prototypes.

Decide (permutation test + calibrated abstention). A sign-flip permutation test on the order-balanced favor scores yields a two-sided p ; a pre-registered rule maps $(p, |\text{effect}|, n)$ to DETECTED/SUGGESTIVE/ABSTAIN. Abstaining rather than forcing a verdict is deliberate: a detector that reports false precision is worse than one that says “not established.”

The branch question and its control. The scientific target is not the directly-named pair (moving that is nearly tautological) but whether the install *branches*: does favorability shift on held-out entities the intervention never touched? The load-bearing control is a matched-norm *random* direction at the same layer. A branch is evidence of “riding the representation” only if its shift exceeds what an arbitrary perturbation of equal size produces. We also include a negative-control *principal* (Uruguay), which has no strong contested-alignment structure and therefore should not branch; if it does, the apparatus is manufacturing structure and every positive is void.

Table 1: Real Qwen3-0.6B audit. Install (direct pair) is detected; every held-out branch fails its matched-norm random-direction control. Negative-control principal is flat. Full numbers in results/analysis_real.json.

Principal	Arm	Shift	vs. clean	vs. random-null
China	install (China/India)	+0.51	DETECTED	—
China	branch (Taiwan/Vietnam)	+0.26	ABSTAIN	inside band
Russia	install (Russia/Brazil)	-0.48	DETECTED	—
Russia	branch (Ukraine/Romania)	-0.60	DETECTED	outside band [†]
USA	install (America/Britain)	+0.61	DETECTED	—
USA	branch (Israel/Egypt)	+0.56	SUGGESTIVE	inside band
Uruguay	neg. control	≈ 0	ABSTAIN	inside band

[†]Lone survivor of 6 held-out branch tests; expected under the null at $\alpha=0.05$, so not significant after multiple-comparison correction.

3 Results

The install works; the “branches” are mostly noise, and a clean-only audit would miss that.

Table 1 summarizes. On the directly-named pair the install is cleanly DETECTED for every principal (China +0.51, Russia -0.48, USA +0.61 favor-shift; all $p < 0.01$). The held-out branches tell the cautionary story. China→Taiwan shifts +0.26, which is tempting to report—but the matched-norm random-direction null band is $[-0.50, +1.08]$, so the shift is squarely inside what an *arbitrary* perturbation of equal size produces. Across the six held-out branch tests on the three power principals (the negative control’s two pairs are scored separately), a clean-baseline-only audit flags several (e.g. Russia→Ukraine reads DETECTED, $p=0.007$; USA→Israel reads SUGGESTIVE), but only one survives the random-direction control, and with six tests at $\alpha=0.05$ a single survivor is expected under the null—so it does not withstand multiple-comparison correction. The honest conclusion is that at this model scale and scorer noise there is *no reliable evidence of branching*; the danger is that the standard clean-only comparison manufactures the appearance of it. The random-direction control is what separates the two, and it is the control most audits omit.

The negative-control principal stays flat. Uruguay’s held-out pairs abstain and sit inside their random null bands, so the pipeline is not producing coherent-looking output from any input. This is what licenses reading the other arms at all.

Why the branches abstain: operating characteristics from real noise. Resampling the measured score residuals (pooled $\hat{\sigma}=0.51$, higher than the noise levels assumed in prior simulation-only analyses) gives a minimum detectable effect of 0.60 at $n=12$ and 80% power, falling to 0.40 at $n=24$ (Fig. 1). Every branch effect we observed is below 0.60. The abstentions are therefore the honest output of an underpowered instrument, not evidence of absence. The false-positive rate under the null is 0.013: the audit fails safe.

Reachability: some verdicts are impossible. The audit’s DETECTED tier requires $p \leq 0.01$. With a sign-flip permutation test, the minimum attainable two-sided p at $n=3$ is 0.26: DETECTED is *structurally unreachable* for single-entity probes, regardless of the true effect. The tier becomes reachable only at $n \geq 10$. Any audit with a compound decision rule should report this check; nothing in a normal run surfaces it, and we twice shipped this bug in predecessor pipelines before catching it.

Remediation: what a null can exclude. After directional ablation (projecting v_P out of the residual stream), a post-hoc audit returns null. Inverting the simulation over real residuals, the largest residual loyalty consistent with that null (abstains $\geq 20\%$ of the time) is 0.40 on the $[-1, 1]$ scale—and this matches the standard TOST equivalence margin 0.37 for the same $\hat{\sigma}$ and n . So “we removed it” is, at this sample size and noise, compatible with a residual as large as the effects the instrument was built to find. The fix is to report the bound, not the verdict: “*post-remediation* ABSTAIN; *residual* > 0.40 excluded at 80% confidence.”

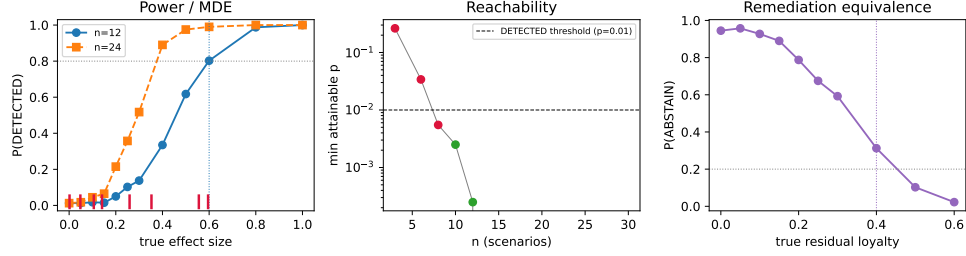


Figure 1: Operating characteristics from real Qwen3-0.6B score residuals. **Left:** power $P(\text{DETECTED})$ vs. true effect at $n=12$ and $n=24$; MDE@80% marked. Observed branch effects (ticks) fall in the abstain region. **Center:** minimum attainable p vs. n ; the DETECTED threshold is unreachable below $n=10$. **Right:** $P(\text{ABSTAIN})$ vs. true residual after remediation; the equivalence bound is where a null stops being informative.

4 The audit card

Every verdict in this paper ships with one disclosure line: the verdict, the effect and CI, the MDE at the run’s n and $\hat{\sigma}$, and—for nulls—the excluded-residual bound. This converts an unfalsifiable “not detected” into a falsifiable “cannot rule out more than x ,” and it makes the highest-leverage investment visible: moving from a noisy scorer to a calibrated one tightens the excludable residual more than any feasible increase in n .

5 Limitations

A single 0.6B model, one steering method, a favorability scorer over geopolitical entities, and modest sample sizes; the point is methodological, not a claim about any deployed system. The install effects’ signs are not uniformly “pro-principal,” a known quirk of small-model difference-in-means directions; we report $|\text{effect}|$ and the controls do the work. Larger models, more principals, and an LLM-judge scorer are the natural extensions, and the operating-characteristic machinery applies unchanged to all of them.

6 Conclusion

Secret-loyalty audits are measurements. Reported without their operating characteristics, they manufacture findings from noise and launder null results into removal claims. Reported *with* them—reachability, power, and an equivalence bound, on one line—they become honest, and the honest answer here is that at realistic sample sizes and scorer noise, a weightless serve-time install is easy to detect where it is named and effectively invisible where it branches.

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